# Amines

**1.** Ethanamine can be obtained if the following compound is heated with $[KOH + Br_2]$

1) Ethanamide
2) Methanamide
3) Propionamide
4) All the above

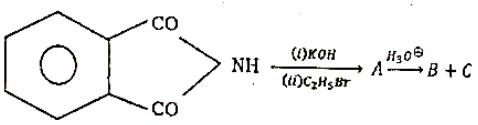
**2.** Gabriel phthalimide reaction is used in the synthesis of

1) Primary aromatic amines
2) Secondary amines
3) Primary aliphatic amines
4) Tertiary amines

**3.** An organic compound A upon reacting with $NH_3$ gives B. On heating, B gives C. C in presence of KOH reacts with $Br_2$ to give $CH_3CH_2NH_2$. A is:-

1) $CH_3COOH$
2) $CH_3CH_2CH_2COOH$
3) $CH_3-\underset{\underset{CH_3}{|}}{CH}-COOH$
4) $CH_3CH_2COOH$

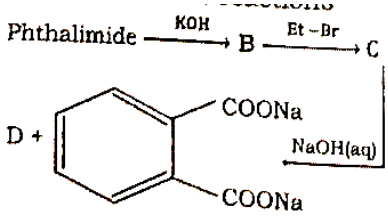
**4.**




B and C in the above sequence are

1) Benzoic acid + aniline
2) Phthalic acid + ethylamine
3) Phthalic acid + aniline
4) Benzoic acid + ethylamine

**5.** In the given set of reactions




D is:

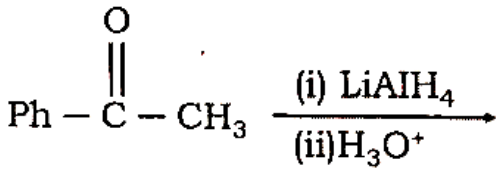
1) $CH_3NH_2$
2) $CH_3CH_2NH_2$
3) $CH_3CH_2NHCH_3$
4) $(CH_3)_2NH$

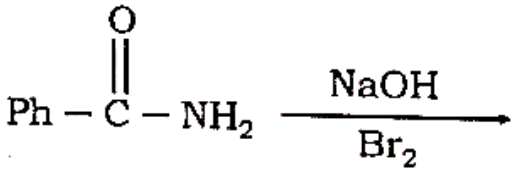
**6.** Method by which Aniline cannot be prepared is

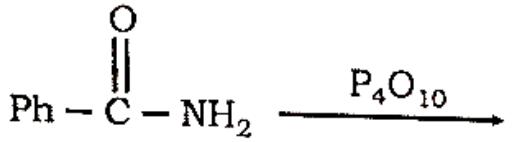
1) Reductio of nitrobenzene with H2/Pd in ethanol.

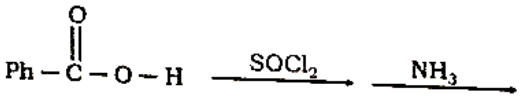
2) Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
3) Hydrolysis of phenylisocyanide with acidic solution
4) Degradation of benzamide with bromine in alkaline solution

**7.** In which of the following reaction cyanide will be obtained as a major product?

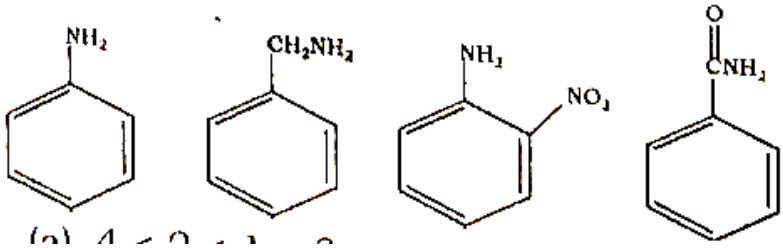
1) 


2) 


3) 

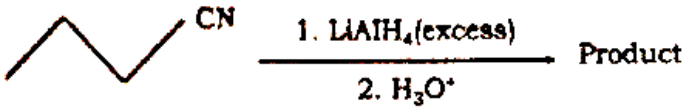

4) 

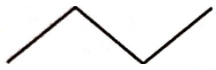

**8.** Rank the following compounds in order of increasing basic strength (weakest →strongest):




1) 4 < 2< 1< 3
2) 4 < 3 < 1 <2
3) 4 < 1 < 3 < 2
4) 2 < 1 < 3 < 4

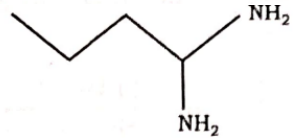
**9.** Choose the appropriate product for this reaction.




1) 

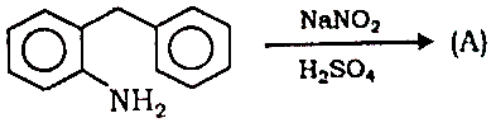
2)


3)

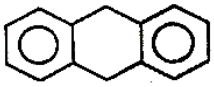
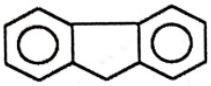
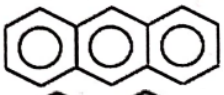
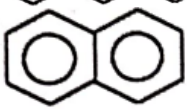

4) 

**10.** A nitrile X is treated with $LiAlH_4$ to obtain compound $Y(C_2H_7N)$. In a separate reaction X is hydrolyzed in an acid medium to obtain Z. The product obtained after mixing Y and Z will be

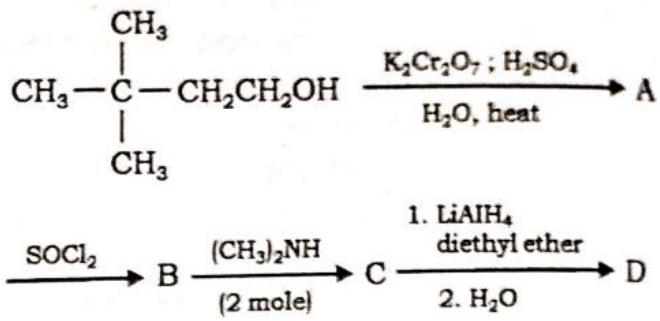
1) $CH_3CONHCH_2CH_3$
2) $CH_3CH_2CONHCH_2CH_3$
3) $(CH_3COO^-)(CH_3CH_2NH_3^+)$
4) $(CH_3CH_2COO^-)(CH_3NH_2^+)$

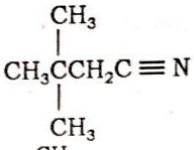
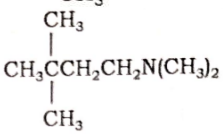
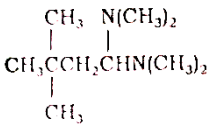
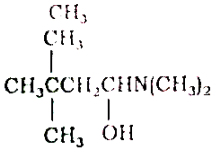



**11.** Product of this reaction is:

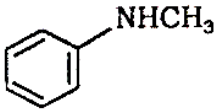
1) 
2) 
3) 
4) 

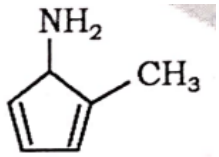
**12.** Identify product D in the following reaction sequence:

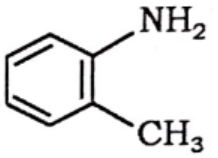


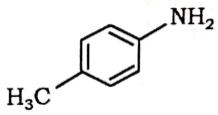

1) 
2) 
3) 
4) 

**13.** The compound $X(C_7H_9N)$ reacts with benzensulfonyl chloride to give $Y(C_{13}H_{13}NO_2S)$, which is insoluble in alkali. The compound X is:

1) 


2) 


3) 


4) 


**14.** Which of the following amine does not react with Hinsberg's reagent?

1) $H_3C-CH_2-NH_2$

2) $H_3C-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$

3) $(CH_3)_3N$

4) $CH_3NH_2$

**15.** Amongst the following, the strongest base in aqueous medium is:

1) $CH_3NH_2$

2) $(CH_3)_3N$

3) $(CH_3)_2NH$

4) $C_6H_5NHCH_3$

**16.** An organic compound (A) on reduction gave a compound (B). Upon treatment with $HNO_2$, (B) gave ethyl alcohol and on warming with $CHCl_3$ and alcoholic KOH, (B) gave an offensive smell. The compound (A) is

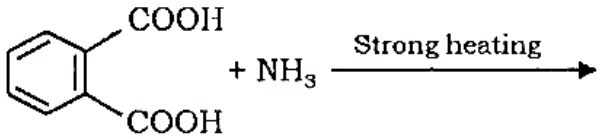
1) $CH_3NH_2$

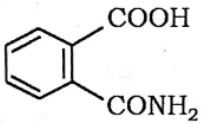
2) $CH_3NC$

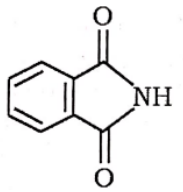
3) $CH_3CN$

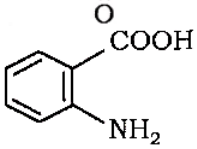
4) $C_2H_5CN$

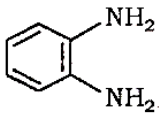
**17.** The major product of the following reaction is:

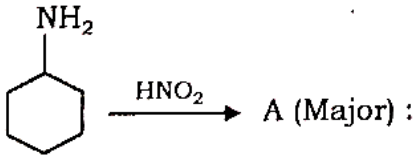



1) 

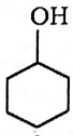
2) 

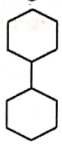

3) 

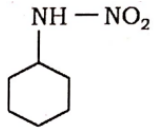

4) 

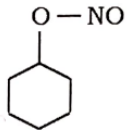




**18.** Product (A) is:

1) 


2) 

3) 


4) 


**19.** Methylamine on treatment with chloroform and ethanolic caustic alkali gives foul smelling compound is

1) $CH_3NCO$
2) $CH_3CNO$
3) $CH_3CN$
4) $CH_3NC$

**20.** Chloroform and ethanolic KOH is used as a reagent in the following reaction :

1. Hoffmann carbylamine reaction
2. Hoffmann degradation reaction
3. Reimer-Tiemann reaction
4. Hoffmann mustard oil reaction

Code is:

1) Only for 1
2) Only for 1 and 2
3) Only for 2 and 4
4) Only for 1 and 3

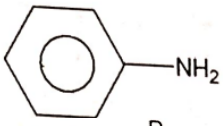
**21.** Aniline on treatment with bromine water yields white precipitate of:

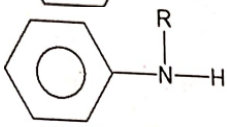
1) o-Bromoaniline
2) p-Bromonailine
3) 2,4,6-Tribromoaniline
4) m-Bromoaniline

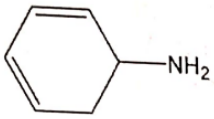
**22.** Which compound does not show diazo reaction:

1) Aniline
2) p-Toluidine
3) p-Nitroaniline
4) Benzylamine

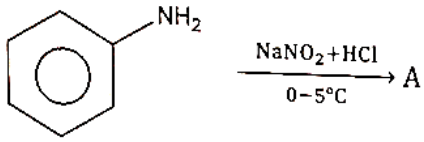
**23.** Which nitroso of the following amines give N-nitroso derivative with $NaNO_2$ and HCl:

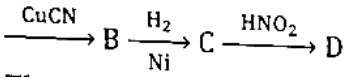
1) $C_2H_5$
2) 

3) 

4) 


**24.** Aniline in a set of reactions yielded end product D

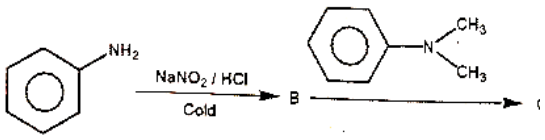


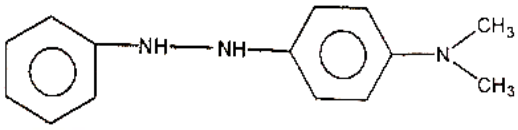

The structure of the product D would be

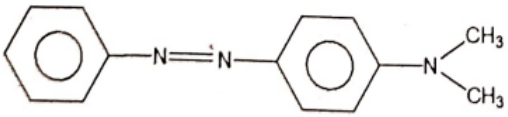
1) $C_6H_5CH_2OH$
2) $C_6H_5CH_2NH_2$
3) $C_6H_5NHOH$
4) $C_6H_5NHCH_2CH_3$

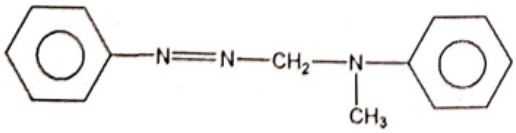
**25.** In a reaction of aniline a coloured product C was obtained.

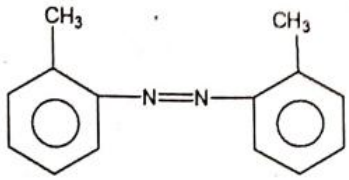



The structure C would be:

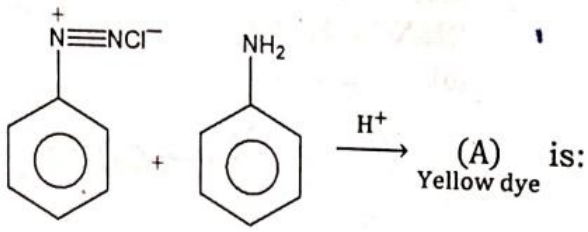
1) 

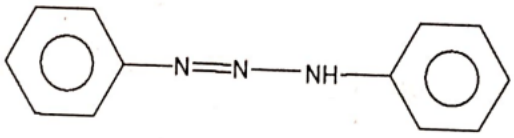
2) 

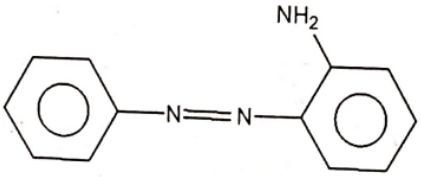
3) 

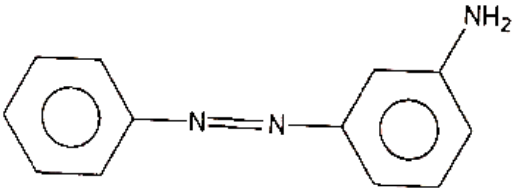
4) 

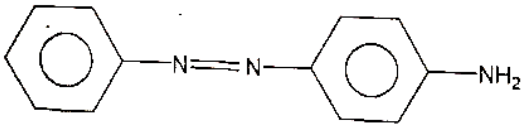
**26.** In the following reaction, the product (A)




1) 

2) 

3) 

4) 

**27.** In the reaction sequence A, B and C are:

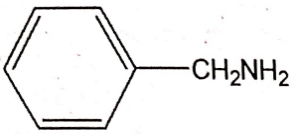
$$A \xrightarrow[HCl]{SnCl_2} B \xrightarrow[0^0C]{NaNO_2/HCl} C \xrightarrow{H_2O/\Delta} C_6H_5OH$$

1) Benzene, nitrobenzene, aniline
2) Nitrobenzene, aniline and azo- compound
3) Nitrobenzene, benzene, aniline
4) Benzene, amino compound, aniline

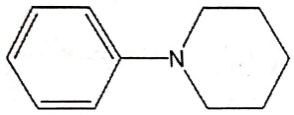
**28.** Reaction of benzenediazonium chloride with alkaline $\beta-$napthol gives an azo dye. This is an example of:

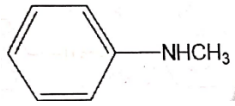
1) Electrophilic substitution
2) Nucleophilic substitution
3) Oxidative coupling
4) A free radical reaction

**29.** Which amine yields N-nitroso amine after treatment with nitrous acid ($NaNO_2$, HCI)?

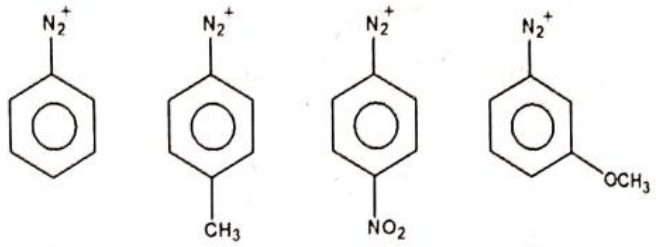
1) 

2)


3) 


4) 


**30.** Write the correct order of the following diazonium cations for the coupling with phenol




I II III IV

1) III > IV > I > II
2) II > I > IV > III
3) III > I > II > IV
4) II > IV > I > III

Answer Key

| | | | |
|---|---|---|---|
| **1** | 3 | **16** | 3 |
| **2** | 3 | **17** | 2 |
| **3** | 4 | **18** | 1 |
| **4** | 2 | **19** | 4 |
| **5** | 2 | **20** | 4 |
| **6** | 2 | **21** | 3 |
| **7** | 3 | **22** | 4 |
| **8** | 2 | **23** | 3 |
| **9** | 2 | **24** | 1 |
| **10** | 3 | **25** | 2 |
| **11** | 2 | **26** | 4 |
| **12** | 2 | **27** | 2 |
| **13** | 1 | **28** | 1 |
| **14** | 3 | **29** | 4 |
| **15** | 3 | **30** | 1 |